

Chapter 2

Intelligent Agents

Learning Objectives

- ☐ Introduction
- ☐ Agents and Environments
- ☐ Acting of Intelligent Agents (Rationality)
- ☐ Structure of Intelligent Agents
- ☐ Agent Types
 - ☐ Simple reflex agent
 - ☐ Model-based reflex agent
 - ☐ Goal-based agent
 - ☐ Utility-based agent
 - ☐ Learning agent
- ☐ Important Concepts and Terms

Introduction

- Intelligent **agents** in AI are autonomous, goal-oriented software or hardware systems that perceive their environment through sensors (or APIs), process data, and take action to achieve specific goals
- often learning from experience to improve performance.
- They range from simple, rule-based systems to complex, adaptive AI, such as autonomous vehicles or personal assistants, which operate independently of human supervision.

What is "Intelligent" in AI?

- In AI, Intelligence isn't about having a soul or "feeling" things.
- It refers to the ability of a system to process information and apply it toward achieving a goal.
- Specifically, it involves:
 - **Learning:** Improving performance based on past data (Experience).
 - **Reasoning:** Using rules to reach nearly certain or approximate conclusions.
 - **Self-Correction:** Refining its own algorithms to get better results over time.
 - **Problem Solving:** Finding a sequence of actions that leads from a current state to a desired goal.

What is an "Agent" in AI?....

- An Agent is any entity that perceives its environment through sensors and acts upon that environment through actuators.
- Think of it as a "doer." An agent doesn't just sit there like a static database; it follows a cycle of Perceive => Think => Act.
 - **Sensors**: Cameras, microphones, keyboard inputs, or data streams.
 - **Actuators**: Wheels, screen displays, speakers, or sending a line of code.

What is an Agent in AI?....

- in AI , **Agent** is any entity that can **perceive its environment through sensors and act upon it using actuators**.
- Think of it as a loop: it sees or senses something, decides what to do, and then acts.
- Here's a simple analogy: a **self-driving car** is a classic example of an AI agent. Its cameras and radar act as **sensors**. Its steering, acceleration, and braking systems act as **actuators**.
- It's constantly analyzing what's around it, other cars, pedestrians, and road signs, and choosing the next best move.
- That loop of sensing, deciding, and acting is what makes it an agent.
- The key idea is: **an AI agent is autonomous**.
- It can make decisions on its own, based on the data it perceives.

Core Characteristics of an Intelligent Agent

Characteristic	Description
Autonomy	It operates without direct intervention from humans.
Reactivity	It perceives changes in the environment and responds in real-time.
Pro-activeness	It doesn't just react; it takes initiative to reach a goal.
Social Ability	It can interact with other agents or humans to complete a task.

Simple Example: A basic thermostat is an **Agent** (it senses temp and turns on heat).

A Nest Learning Thermostat is an **Intelligent Agent** (it learns your schedule, senses if you're home, and optimizes for energy savings automatically)

Agents and Environments

- To understand how AI works in the real world, you have to look at the relationship between the **Agent** (the "brain" and "body") and the **Environment** (the world it lives in).
- The Agent exists in a loop: it takes in data from the environment, processes it, and then spits an action back out to change that environment.

Agent-Environment Loop

- In AI theory, this relationship is often expressed through the **PEAS** framework. When designing an Intelligent Agent, engineers define these four pillars:
 - **P – Performance Measure:** The criteria for success (e.g., "How clean is the floor?" or "How fast did the car get to its destination?").
 - **E – Environment:** The "playing field" (e.g., a living room, a highway, or a stock market).
 - **A – Actuators:** The tools used to act (e.g., motors, speakers, or screen pixels).
 - **S – Sensors:** The tools used to perceive (e.g., cameras, GPS, or LiDAR).

Types of Environments

The category of environments based on several key dimensions:

Dimension	Description	Example
Fully vs. Partially Observable	Can the agent see everything at once?	Chess (Fully) vs. Poker (Partially - hidden cards)
Deterministic vs. Stochastic	Is the next state certain or is there randomness?	Calculators (Deterministic) vs. Weather (Stochastic)
Static vs. Dynamic	Does the world change while the agent is "thinking"?	Crossword Puzzle (Static) vs. Driving (Dynamic)
Discrete vs. Continuous	Are there a limited number of actions/states?	Chess (Discrete squares) vs. Robot Arm (Continuous motion)
Single vs. Multi-agent	Is the agent alone or competing/cooperating?	Solitaire (Single) vs. Online Gaming (Multi)

How Agents Process the Environment

- ✓ Depending on how complex the environment is, we design different "brains" for the agent:
- ✓ **Simple Reflex Agents:** These follow "If-Then" rules. They don't remember the past; they only care about the current moment. (Example: A motion-sensor light).
- ✓ **Model-Based Agents:** These keep an internal "map" of the parts of the environment they can't see right now. They have a "memory."
- ✓ **Goal-Based Agents:** These act specifically to reach a target. They look ahead and ask, "Will this action get me closer to my goal?"
- ✓ **Utility-Based Agents:** These don't just want to reach a goal; they want to reach it in the *best* way possible (e.g., the fastest, cheapest, or safest route).

intelligent agent.....

- An **intelligent agent** is an entity that:
 - ✓ **Perceives** its environment through sensors and
 - ✓ **Acts** upon the environment through actuators

Example:

- A **human** : eyes (sensors), hands/legs (actuators)
- A **robot** : cameras (sensors), motors (actuators)
- A **software agent**: input data (sensors), outputs/actions (actuators)

"Rationality"

- A common mistake is thinking an Intelligent Agent must be "perfect."
- Actually, it just needs to be **Rational**. A rational agent chooses the action that is expected to maximize its performance measure, based on the evidence it has gathered from its sensors.
- Even if it fails, it can still be "intelligent" if it made the best possible choice with the info it had.

What is Rationality?

- In AI, rationality means Doing the right action based on what the agent knows, to achieve the best expected outcome.
- A rational agent:
 - Chooses actions that maximize its performance measure
 - Uses the information it has, not perfect information
 - Considers uncertainty and future consequences

Important:

- Rational $\neq$ Perfect
- Rational $\neq$ Always successful

Acting of Intelligent Agents (Rationality)

- **Acting rationally** is a core idea in Artificial Intelligence. It explains *how* an intelligent agent should choose its actions.
- In AI, **Rationality** is the gold standard for how an agent "acts." It isn't about being "human-like" or having common sense; it is a mathematical and logical commitment to making the best possible decision given the circumstances.
- A rational agent is one that acts so as to achieve the **best outcome** or, when there is uncertainty, the best **expected** outcome.

What Makes an Action "Rational"?

- Rationality is not defined by the result (success or failure), but by the **quality of the decision-making process**. An agent's rationality depends on four factors:
 - 1. The Performance Measure:** The criteria that define success (the "goal").
 - 2. Prior Knowledge:** What the agent already knows about the world.
 - 3. Percept Sequence:** Everything the agent has sensed or experienced up to this point.
 - 4. Action Choices:** The set of things the agent is actually capable of doing.

Rationality vs. Perfection

- This is a crucial distinction in AI.
- **Perfection (Omniscience)** implies knowing the actual outcome of an action and never making a mistake.
- This is impossible in the real world because the future is often unpredictable.
- **Rationality** implies making the best choice with the information available.
- **The "Crossing the Street" Example:**
- If you look both ways, see no cars, and start crossing but a piano falls on you from a window your action was still rational (based on what you could sense), even though the outcome was not perfect.

Components of Rational Acting

- To act rationally, an agent typically follows these three behaviors:

1) Information Gathering (Exploration)

- A rational agent doesn't just act blindly. Sometimes, the most rational action is to stop and learn.
- This is called **Exploration**. For example, a vacuum robot might spend its first five minutes just bumping into walls to "map" the room so it can be more efficient later.

Cont...

2) Learning

- A rational agent doesn't just rely on its "factory settings." It modifies its future behavior based on its past perceptions. This is called **Autonomy**. An agent that ignores its own experience is not being rational.

3) Expected Utility

- In complex environments, an agent rarely has a 100% guarantee of success. Therefore, it calculates the Expected Utility (EU) of its actions:

Structure of Intelligent Agents

- The structure of an agent in AI is the foundation of how it works. It's not just about what the agent *does*, but **how** it processes the world around it to take action. At its core, every agent has a few essential components:

1. Sensors

- These are the agent's eyes and ears, what it uses to perceive the environment. Depending on the system, sensors could be:
 - Cameras in a robot
 - Microphones in a voice assistant
 - Temperature detectors in a smart thermostat
 - Web data inputs into a recommendation engine
- Sensors gather raw data about the state of the world.

Cont..

- **2. Actuators**
- These are the parts that let the agent act. In physical systems, they could be motors, servos, or mechanical parts.
- In software systems, actions might include sending alerts, modifying data, or triggering other digital processes.
- If a thermostat senses that a room is too cold, the actuator is what actually **turns on the heater**.

3. Percept Sequence

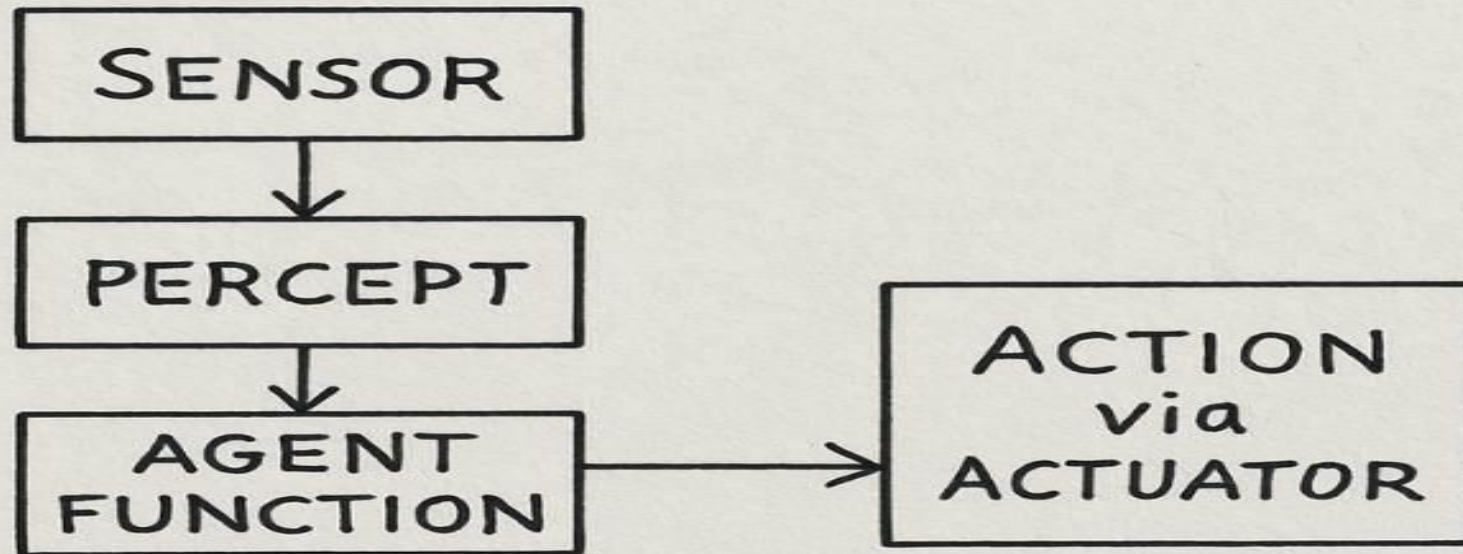
- This is essentially the **history of everything the agent has perceived so far**. Rather than working on just the latest data point, some agents need to analyze trends over time, recognize patterns, or understand cause-effect relationships that emerge across multiple events.
- Not every agent uses its full percept sequence, but more complex ones definitely do.

4. Agent Function

- This is the real decision-maker. The agent function takes the percept (or percept sequence) and maps it to a specific action. It's the logic or strategy that says, "Given what I'm sensing, what should I do?"
- You could think of the agent function as the *brain* of the operation, though it can range from something super simple (like a rule-based system) to something highly complex and adaptive (like a deep learning model).

Cont....

STRUCTURE OF AN AI AGENT



Agent Program vs Agent Function

These two often get mixed up, but they're not the same.

- **Agent Function** is the **idealized mathematical model**. It's like the full rulebook of what the agent *should* do in every possible situation, even if that's not feasible in practice.
- **Agent Program** is the **actual implementation**. It's the code that gets written to carry out the logic of the agent function in a real system, with real constraints.
- Think of it like this: a **recipe** is the function; the **robot chef** that follows it is the program. And sometimes, the robot needs to tweak things if an ingredient is missing or the oven's acting up.

Example: Smart Thermostat

Let's put it all together.

- **Sensors:** temperature sensor detects 18°C
- **Percept Sequence:** logs that it's been under 20°C for 2 hours
- **Agent Function:** determines the optimal comfort range is 21-23°C
- **Actuator:** sends a signal to the heater to turn on
- Here, the thermostat is making a decision based on environmental input, a memory of recent readings, and a built-in rule set. It doesn't need a fancy neural network to be an intelligent agent; it just needs the structure to perceive, decide, and act.

Types of AI Agents?

- This is where it gets interesting. Not all AI agents are built the same.
- Depending on the complexity of the environment and the task at hand, different types of agents are used.
- There are **five primary types** of AI agents and what makes them unique.

1. Simple Reflex Agent?

What is a [Simple Reflex Agent](#)?

- These agents operate on IF-THEN rules. They don't remember past actions or look ahead. They just act based on the current situation.
- Behavior: “If I see dirt, start vacuuming.”
- No memory or planning
- Only responds to current percept
- Example: A basic vacuum cleaner robot. It detects dirt → it moves. That's it. If it hits a wall, it turns.
- Simple? Yes. Effective in limited scenarios? Absolutely.

2. Model-Based Reflex Agent?

What is a Model-Based Reflex Agent?

- These agents are a step up. They maintain some kind of internal state that represents part of the world they can't directly sense at the moment.
- This means they can handle situations where not everything is visible all the time.
- Keeps a model of the world
- Updates internal state as new data comes in
- Example: A home security system. If a window sensor is triggered and no one disarmed the alarm, it assumes a break-in. It remembers past states (armed, disarmed, motion detected) and acts accordingly.

3. Goal-Based Agent?

What is a Goal-Based Agent?

- Here's where decision-making gets more sophisticated. A goal-based agent doesn't just react; it evaluates whether an action will help it achieve a goal.
- Plans ahead
- Considers different outcomes
- Chooses based on what gets it closer to its goal
- Example: A chess-playing AI. It doesn't just move a piece because it can, it simulates potential future moves and picks the one most likely to lead to a win.
- These agents require more computing power but offer far more intelligent behavior.

4. Utility-Based Agent?

- While goal-based agents focus on reaching a *specific* outcome, utility-based agents go a level deeper. They're not just trying to “win the game”, they want to win in the best possible way.
- A utility-based agent doesn't just ask “*Does this action achieve my goal?*” It asks, “*How good is this outcome compared to all other possible outcomes?*”
 - Evaluates multiple outcomes
 - Uses a utility function to rank them
 - Chooses the action with the highest utility

Example: A self-driving car.

- Let's say the goal is to reach a destination. That could be done by taking the shortest route, but what if that road is congested or has a high accident rate?
- A utility-based agent factors in safety, speed, fuel efficiency, and maybe even passenger comfort, and picks the best route accordingly.
- It's about maximizing value, not just hitting a target.

5. Learning Agent?

- A **learning agent** is capable of improving its behavior over time by **learning from experience**. Instead of being pre-programmed with fixed rules, it adjusts its strategies based on feedback.
- These agents typically have four components:
 1. **Learning Element** – This is what makes improvements based on performance.
 2. **Performance Element** – Responsible for selecting actions.
 3. **Critic** – Gives feedback about how the agent is doing relative to a standard.
 4. **Problem Generator** – Suggests exploratory actions to improve learning.

Example:

- Think of how **recommendation engines** like Netflix or YouTube work. They learn from what you watch, how long you watch, when you skip, and start adjusting what they show you next. That's learning in action.
- These systems are especially useful in environments where things change over time or where the full set of rules isn't known upfront.

Summary Concepts and Terms

- Agent = Architecture + Agent Program
 - Architecture: Hardware/software platform
 - Agent Program: Maps percepts to actions
- Environment $\rightarrow$ Sensors $\rightarrow$ Agent $\rightarrow$ Actuators $\rightarrow$ Environment
- Percept: Input from sensors.
- Actuator: Output mechanism.
- Utility Function: Quantifies preferences.
- Learning Element: Modifies agent for better performance.
- Examples and Case Studies
- Self-Driving Car: Utility-based with learning for traffic prediction.

Question

1. What is rationality in AI?

- Answer: Rationality is the ability of an agent to choose actions that maximize expected performance based on its percepts, knowledge, and available actions.

2. What is the structure of an intelligent agent?

Answer: The structure of an intelligent agent consists of:

- Sensors
- Agent program
- Architecture
- Actuators

- **Exercises:**

Lab: Implement a simple reflex agent in Python (e.g., maze navigation).

This is all about chapter 2